

Overview of materials for Polyester Thermoplastic Elastomer

Categories: [Polymer](#); [Thermoplastic](#); [Elastomer, TPE](#); [Polyester, TP](#); [Polyester Thermoplastic Elastomer](#)

Material Notes:

This property data is a summary of similar materials in the MatWeb database for the category "Polyester Thermoplastic Elastomer". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors:

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Physical Properties	Metric	English	Comments
Density	1.00 - 1.36 g/cc	0.0361 - 0.0491 lb/in³	Average value: 1.19 g/cc Grade Count:148
Water Absorption	0.100 - 0.800 %	0.100 - 0.800 %	Average value: 0.500 % Grade Count:42
Moisture Absorption at Equilibrium	0.200 - 0.560 %	0.200 - 0.560 %	Average value: 0.388 % Grade Count:31
Water Absorption at Saturation	0.500 - 0.900 %	0.500 - 0.900 %	Average value: 0.660 % Grade Count:5
Viscosity	400 - 461000 cP	400 - 461000 cP	Average value: 204000 cP Grade Count:4
 Viscosity	200 - 85000 cP @Temperature 160 - 232 °C	200 - 85000 cP @Temperature 320 - 450 °F	Average value: 43100 cP Grade Count:5
Viscosity Test	68.0 - 150 cm³/g	0.680 - 1.50 dl/g	Average value: 109 cm³/g Grade Count:3
Molecular Weight	458 g/mol	458 g/mol	Average value: 458 g/mol Grade Count:21
Maximum Moisture Content	0.0100 - 0.0500	0.0100 - 0.0500	Average value: 0.0214 Grade Count:51
Linear Mold Shrinkage	0.00400 - 0.0220 cm/cm	0.00400 - 0.0220 in/in	Average value: 0.0123 cm/cm Grade Count:75
Linear Mold Shrinkage, Transverse	0.0100 - 0.0220 cm/cm	0.0100 - 0.0220 in/in	Average value: 0.0176 cm/cm Grade Count:11
Melt Flow	0.500 - 300 g/10 min	0.500 - 300 g/10 min	Average value: 21.8 g/10 min Grade Count:126
Chemical Properties	Metric	English	Comments
Acid Value	2.00 - 3.00	2.00 - 3.00	Average value: 2.83 Grade Count:6
Mechanical Properties	Metric	English	Comments
Hardness, Shore A	65.0 - 99.0	65.0 - 99.0	Average value: 90.2 Grade Count:55

Hardness, Shore D	19.0 - 78.0	19.0 - 78.0	Average value: 48.9 Grade Count:123
Tensile Strength, Ultimate	2.00 - 56.5 MPa	290 - 8200 psi	Average value: 28.5 MPa Grade Count:116
Tensile Strength, Yield	2.00 - 72.2 MPa	290 - 10500 psi	Average value: 32.9 MPa Grade Count:40
Elongation at Break	8.00 - 2000 %	8.00 - 2000 %	Average value: 496 % Grade Count:142
Elongation at Yield	6.00 - 44.0 %	6.00 - 44.0 %	Average value: 22.1 % Grade Count:8
Modulus of Elasticity	0.0130 - 2.25 GPa	1.89 - 326 ksi	Average value: 0.345 GPa Grade Count:37
	0.0466692 - 3.14966 GPa @Temperature -100 - 110 °C	6.76890 - 456.827 ksi @Temperature -148 - 230 °F	Average value: 1.26 GPa Grade Count:2
Flexural Yield Strength	1.00 - 89.6 MPa	145 - 13000 psi	Average value: 20.0 MPa Grade Count:28
Flexural Modulus	0.0130 - 2.76 GPa	1.89 - 400 ksi	Average value: 0.369 GPa Grade Count:116
	0.00800 - 2.50 GPa @Temperature -40.0 - 100 °C	1.16 - 363 ksi @Temperature -40.0 - 212 °F	Average value: 0.758 GPa Grade Count:10
Compressive Modulus	0.00240 - 0.0392 GPa	0.348 - 5.69 ksi	Average value: 0.0153 GPa Grade Count:29
Resilience	40.0 - 81.0	40.0 - 81.0	Average value: 61.4 Grade Count:36
Flex Crack Resistance	1.00e+6	1.00e+6	Average value: 1.00e+6 Grade Count:7
Izod Impact, Notched	0.374 J/cm - NB	0.700 ft-lb/in - NB	Average value: 4.24 J/cm Grade Count:75
	0.26695 J/cm - NB @Temperature -40.0 - -40.0 °C	0.50011 ft-lb/in - NB @Temperature -40.0 - -40.0 °F	Average value: 1.10 J/cm Grade Count:8
	0.294199 - 980.567 J/cm @Temperature -40.0 - -40.0 °C	0.551156 - 1837.00 ft-lb/in @Temperature -40.0 - -40.0 °F	Average value: 1.10 J/cm Grade Count:29
	0.294199 - 980.567 J/cm @Thickness 6.40 - 6.40 mm	0.551156 - 1837.00 ft-lb/in @Thickness 0.252 - 0.252 in	Average value: 1.10 J/cm Grade Count:29
Izod Impact, Unnotched	13.3 - 5340 J/cm	25.0 - 10000 ft-lb/in	Average value: 14.7 J/cm Grade Count:5
Izod Impact, Notched (ISO)	4.70 - 21000 kJ/m²	2.24 - 9990 ft-lb/in²	Average value: 20.9 kJ/m² Grade Count:16
	3.50 - 20997.9 kJ/m² @Temperature -40.0 - -20.0 °C	1.67 - 9991.75 ft-lb/in² @Temperature -40.0 - -4.00 °F	Average value: 19.9 kJ/m² Grade Count:18
Charpy Impact Unnotched	7.10 J/cm² - NB	33.8 ft-lb/in² - NB	Average value: 8.65 J/cm² Grade Count:26
	0.450 J/cm² - NB @Temperature -40.0 - -30.0 °C	2.14 ft-lb/in² - NB @Temperature -40.0 - -22.0 °F	Average value: 0.450 J/cm² Grade Count:20
Charpy Impact, Notched	0.940 J/cm² - NB	4.47 ft-lb/in² - NB	Average value: 4.98 J/cm² Grade Count:38
	0.450 J/cm² - NB @Temperature -40.0 - -30.0 °C	2.14 ft-lb/in² - NB @Temperature -40.0 - -22.0 °F	Average value: 5.20 J/cm² Grade Count:28
Gardner Impact	>= 32.7657 J @Temperature -40.0 - -40.0 °C	>= 24.1667 ft-lb @Temperature -40.0 - -40.0 °F	Average value: 49.4 J Grade Count:2
Tensile Creep Modulus, 1 hour 	30.7907 - 111.849 MPa @Temperature 40.0 - 100 °C	4465.82 - 16222.4 psi @Temperature 104 - 212 °F	Average value: 66.3 MPa Grade Count:2

	30.7907 - 111.849 MPa @Pressure 0.500 - 7.00 MPa	4465.82 - 16222.4 psi @Pressure 72.5 - 1020 psi	Average value: 66.3 MPa Grade Count:2
Tensile Creep Modulus, 1000 hours 	21.7457 - 83.3376 MPa @Temperature 40.0 - 100 °C	3153.95 - 12087.1 psi @Temperature 104 - 212 °F	Average value: 49.2 MPa Grade Count:2
	21.7457 - 83.3376 MPa @Pressure 0.500 - 7.00 MPa	3153.95 - 12087.1 psi @Pressure 72.5 - 1020 psi	Average value: 49.2 MPa Grade Count:2
Tear Strength	40.0 - 362 kN/m	228 - 2070 pli	Average value: 149 kN/m Grade Count:83
Taber Abrasion, mg/1000 Cycles	5.00 - 90.0	5.00 - 90.0	Average value: 20.7 Grade Count:33
Compression Set	1.00 - 65.0 %	1.00 - 65.0 %	Average value: 42.9 % Grade Count:37
Torsional Stiffness	20.0 - 500 MPa	2900 - 72500 psi	Average value: 198 MPa Grade Count:11
Electrical Properties	Metric	English	Comments
Electrical Resistivity	1.00e+11 - 4.00e+16 ohm-cm	1.00e+11 - 4.00e+16 ohm-cm	Average value: 1.21e+15 ohm-cm Grade Count:59
Surface Resistance	1.00e+12 - 2.00e+17 ohm	1.00e+12 - 2.00e+17 ohm	Average value: 2.66e+16 ohm Grade Count:16
Dielectric Constant	3.30 - 5.10	3.30 - 5.10	Average value: 4.47 Grade Count:12
Dielectric Strength	11.8 - 75.0 kV/mm	300 - 1910 kV/in	Average value: 23.1 kV/mm Grade Count:59
Dissipation Factor	0.0200 - 0.0400	0.0200 - 0.0400	Average value: 0.0336 Grade Count:11
Arc Resistance	120 - 240 sec	120 - 240 sec	Average value: 165 sec Grade Count:4
Comparative Tracking Index	100 - 600 V	100 - 600 V	Average value: 590 V Grade Count:40
Hot Wire Ignition, HWI	15.0 - 120 sec	15.0 - 120 sec	Average value: 47.8 sec Grade Count:4
High Amp Arc Ignition, HAI	120 arcs	120 arcs	Average value: 120 arcs Grade Count:4
Thermal Properties	Metric	English	Comments
CTE, linear	23.4 - 250 µm/m-°C	13.0 - 139 µin/in-°F	Average value: 169 µm/m-°C Grade Count:45
CTE, linear, Transverse to Flow	170 - 190 µm/m-°C	94.4 - 106 µin/in-°F	Average value: 183 µm/m-°C Grade Count:3
Melting Point	83.0 - 223 °C	181 - 433 °F	Average value: 194 °C Grade Count:121
Maximum Service Temperature, Air	68.0 - 161 °C	154 - 322 °F	Average value: 113 °C Grade Count:19

Deflection Temperature at 0.46 MPa (66 psi)	42.0 - 145 °C	108 - 293 °F	Average value: 84.4 °C Grade Count:37
Deflection Temperature at 1.8 MPa (264 psi)	45.0 - 130 °C	113 - 266 °F	Average value: 62.9 °C Grade Count:13
Vicat Softening Point	42.0 - 210 °C	108 - 410 °F	Average value: 151 °C Grade Count:62
Softening Point	92.0 - 166 °C	198 - 331 °F	Average value: 131 °C Grade Count:5
Brittleness Temperature	-78.0 - -50.0 °C	-108 - -58.0 °F	Average value: -64.3 °C Grade Count:32
Glass Transition Temp, Tg	-65.0 - 5.50 °C	-85.0 - 41.9 °F	Average value: -30.8 °C Grade Count:7
UL RTI, Electrical	50.0 °C	122 °F	Average value: 50.0 °C Grade Count:6
UL RTI, Mechanical with Impact	50.0 °C	122 °F	Average value: 50.0 °C Grade Count:6
UL RTI, Mechanical without Impact	50.0 °C	122 °F	Average value: 50.0 °C Grade Count:6
Flammability, UL94	HB - V-0	HB - V-0	Grade Count:56
Oxygen Index	20.0 - 37.2 %	20.0 - 37.2 %	Average value: 29.8 % Grade Count:3

Processing Properties	Metric	English	Comments
Processing Temperature	20.0 - 300 °C	68.0 - 572 °F	Average value: 137 °C Grade Count:23
Nozzle Temperature	170 - 271 °C	338 - 520 °F	Average value: 222 °C Grade Count:38
Adapter Temperature	180 - 260 °C	356 - 500 °F	Average value: 224 °C Grade Count:27
Die Temperature	170 - 260 °C	338 - 500 °F	Average value: 220 °C Grade Count:37
Melt Temperature	150 - 290 °C	302 - 554 °F	Average value: 215 °C Grade Count:66
Mold Temperature	10.0 - 93.3 °C	50.0 - 200 °F	Average value: 38.8 °C Grade Count:53
Drying Temperature	80.0 - 127 °C	176 - 260 °F	Average value: 93.1 °C Grade Count:68
Moisture Content	0.0100 - 0.150 %	0.0100 - 0.150 %	Average value: 0.0492 % Grade Count:13
Injection Pressure	4.96 - 103 MPa	720 - 15000 psi	Average value: 59.2 MPa Grade Count:5
Cure Time	0.0333 - 0.250 min	0.000556 - 0.00417 hour	Average value: 0.118 min Grade Count:4

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.